Kernix 0.9.0 / Performance brief

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

25 workloads

100 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

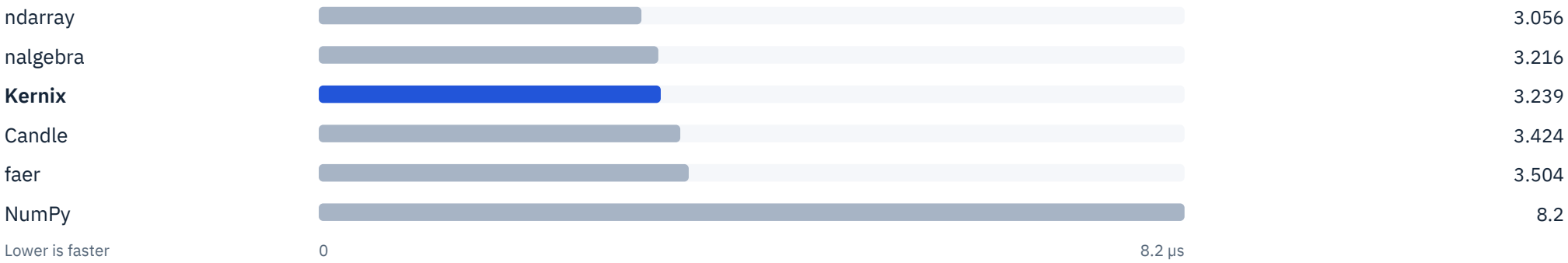
Kernix 0.9.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



Kernix 0.9.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

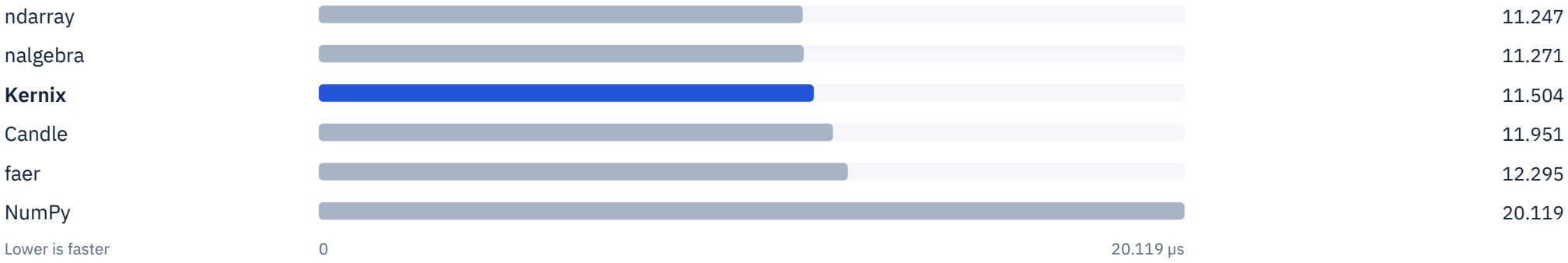
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



Kernix 0.9.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



Kernix 0.9.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



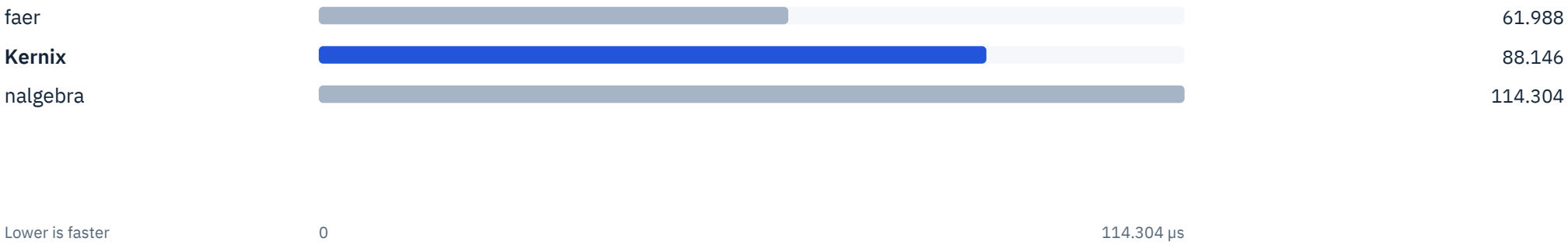
Kernix 0.9.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



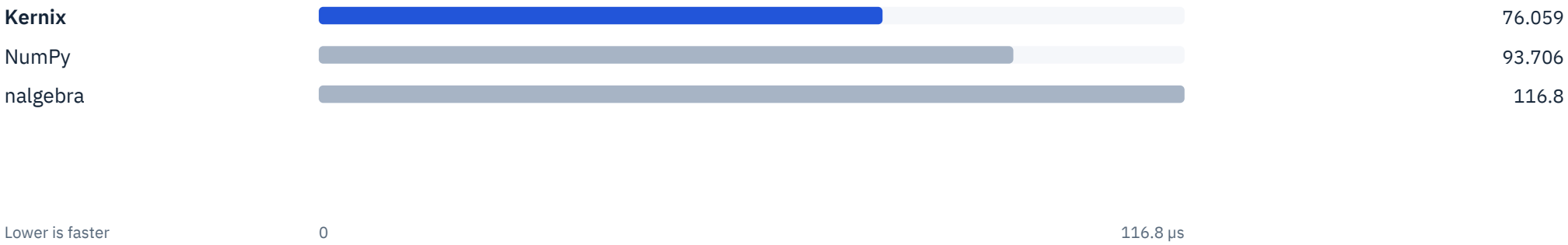
Kernix 0.9.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

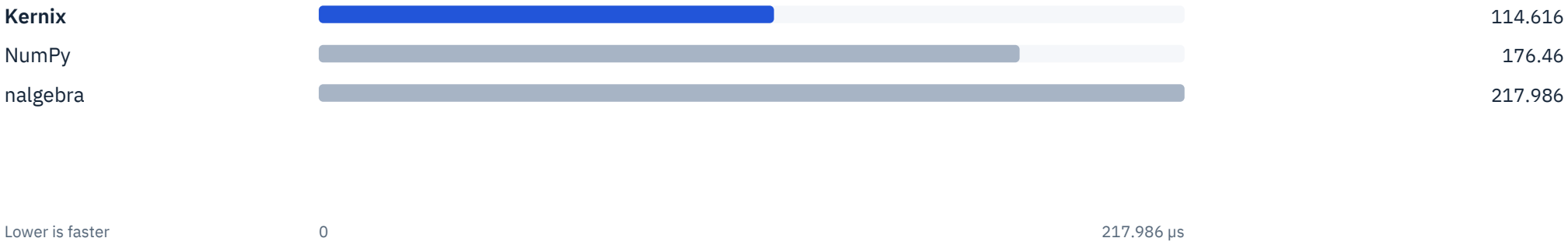
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Kernix 0.9.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



Kernix 0.9.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

128 × 128 · f64 · 1 worker · allocated output



Kernix 0.9.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.9.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Record + backward

Per-call median

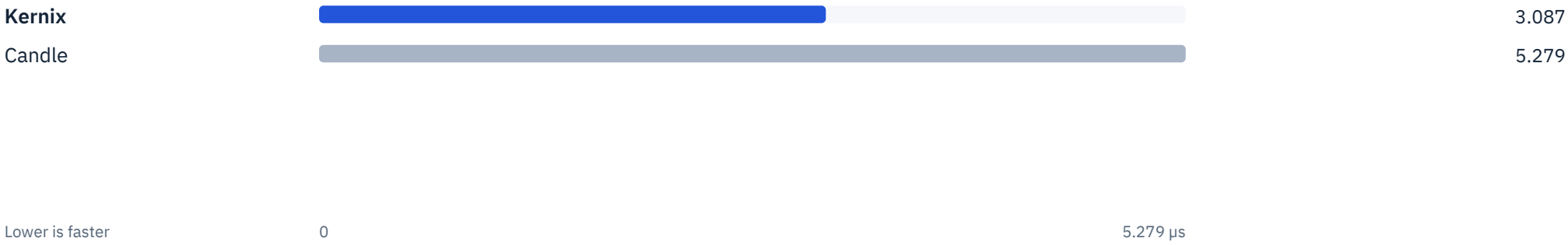
sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



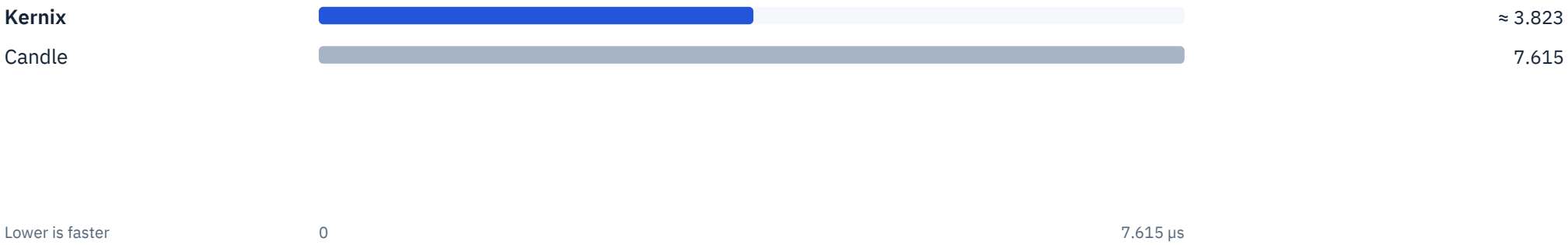
Kernix 0.9.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Model recording + forward

Per-call median

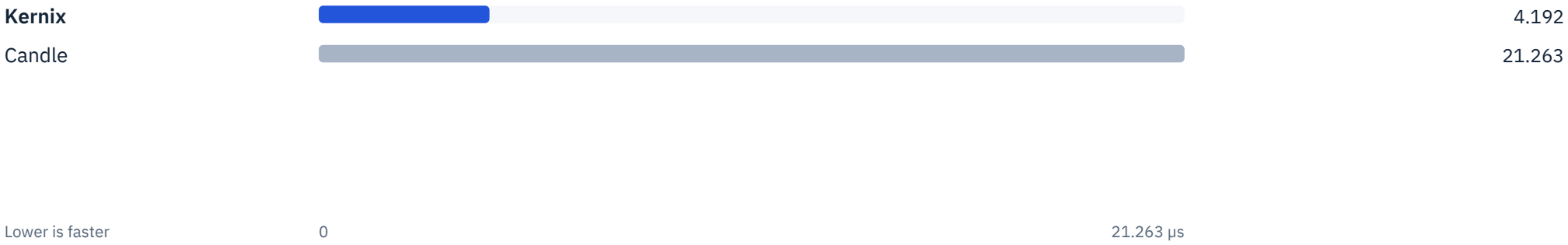
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Kernix 0.9.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Parameter update

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

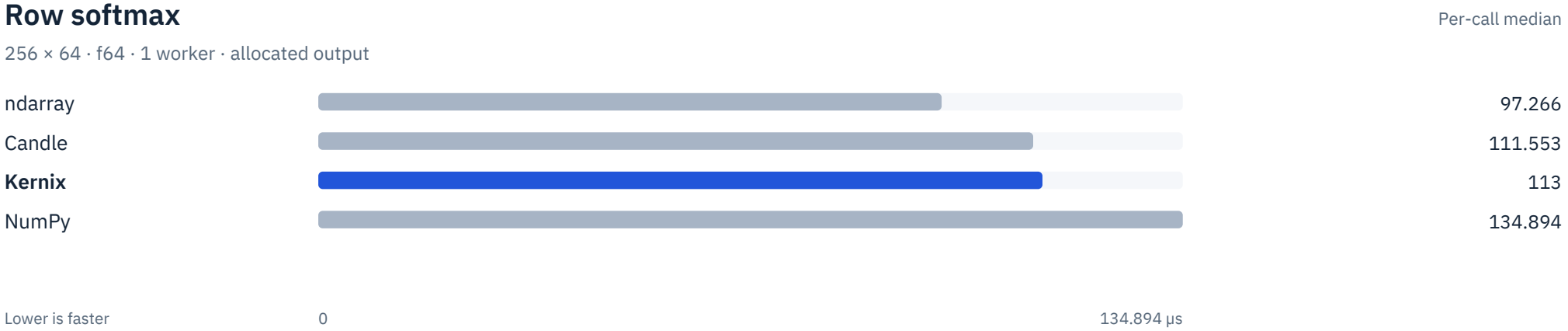


Kernix 0.9.0 / Statistics & activations

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

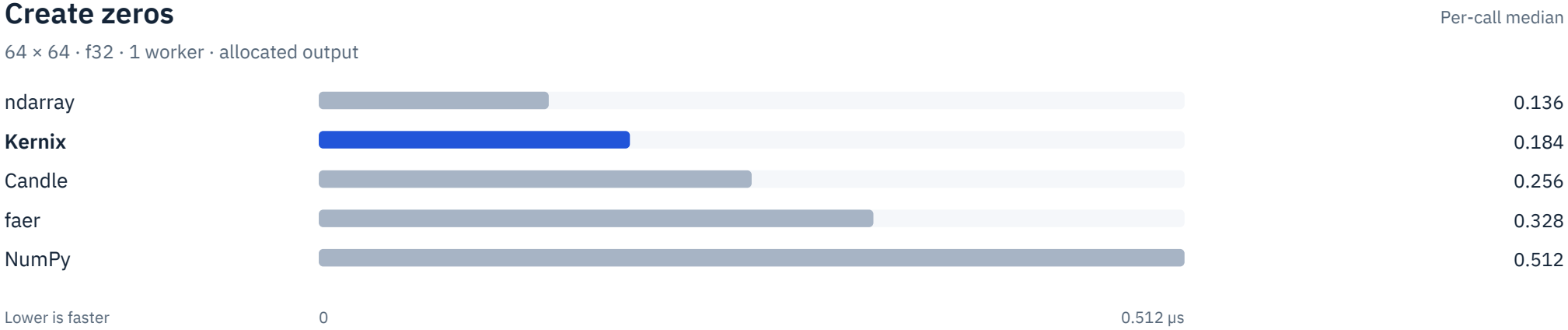


Kernix 0.9.0 / Tensor construction

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output



Create ones

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	3239	3.239
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	3504	3.504
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	3056	3.056
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	3424	3.424
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	8200	8.2
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	3216	3.216
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	31118	31.118
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	22463	22.463
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	22174	22.174
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	21799	21.799

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	22631	22.631
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	22574	22.574
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	169821	169.821
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	196011	196.011
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	208642	208.642
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	207458	207.458
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	210410	210.41
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	169268	169.268
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	1334373	1334.373
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	1466035	1466.035

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	1335542	1335.542
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	1324023	1324.023
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	1548982	1548.982
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	1326502	1326.502
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	11271	11.271
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	20119	20.119
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	11951	11.951
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	11504	11.504
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	12295	12.295
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	11247	11.247

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	94506	94.506
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	100065	100.065
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	94962	94.962
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	85819	85.819
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	93594	93.594
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	87051	87.051
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	729143	729.143
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	674018	674.018
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	770612	770.612
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	761412	761.412

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	757933	757.933
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	765316	765.316
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	4448	4.448
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	12975	12.975
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	3856	3.856
Column-pivoted QR / faer 96 × 32 · f64	61988	61.988
Column-pivoted QR / Kernix 96 × 32 · f64	88146	88.146
Column-pivoted QR / nalgebra 96 × 32 · f64	114304	114.304
Column-pivoted QR / nalgebra 256 × 64 · f64	1151545	1151.545
Column-pivoted QR / faer 256 × 64 · f64	474936	474.936

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / Kernix 256 × 64 · f64	786763	786.763
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	76059	76.059
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	93706	93.706
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	116800	116.8
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	114616	114.616
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	176460	176.46
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	217986	217.986
Thin QR / faer 96 × 32 · f64	65356	65.356
Thin QR / NumPy 96 × 32 · f64	65963	65.963
Thin QR / nalgebra 96 × 32 · f64	61388	61.388

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / Kernix 96 × 32 · f64	70155	70.155
Thin QR / faer 256 × 64 · f64	443554	443.554
Thin QR / Kernix 256 × 64 · f64	690305	690.305
Thin QR / nalgebra 256 × 64 · f64	588263	588.263
Thin QR / NumPy 256 × 64 · f64	576105	576.105
Cholesky factorization / Kernix 64 × 64 · f64	27334	27.334
Cholesky factorization / nalgebra 64 × 64 · f64	20350	20.35
Cholesky factorization / faer 64 × 64 · f64	13256	13.256
Cholesky factorization / NumPy 64 × 64 · f64	18031	18.031
Cholesky factorization / faer 128 × 128 · f64	64996	64.996

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 128 × 128 · f64	280789	280.789
Cholesky factorization / NumPy 128 × 128 · f64	82746	82.746
Cholesky factorization / nalgebra 128 × 128 · f64	128775	128.775
Cholesky factorization / nalgebra 256 × 256 · f64	1031706.0000000001	1031.7060000000001
Cholesky factorization / faer 256 × 256 · f64	407037	407.037
Cholesky factorization / Kernix 256 × 256 · f64	1994912.0000000002	1994.9120000000002
Cholesky factorization / NumPy 256 × 256 · f64	751805	751.805
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3512	3.512
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	7128	7.128
Model inference / Kernix batched normalized/sliced MLP · f64	3087	3.087

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model inference / Candle batched normalized/sliced MLP · f64	5279	5.279
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3822.9999999999995	3.8229999999999995
Model recording + forward / Candle batched normalized/sliced MLP · f64	7615	7.615
Model backward / Candle prepared graph → fresh gradients · f64	21263	21.263
Model backward / Kernix prepared graph → fresh gradients · f64	4192	4.192
Parameter update / Kernix transactional SGD, fixed gradients · f64	736	0.736
Row softmax / ndarray 256 × 64 · f64	97266	97.266
Row softmax / NumPy 256 × 64 · f64	134894	134.894
Row softmax / Candle 256 × 64 · f64	111553	111.553
Row softmax / Kernix 256 × 64 · f64	113000	113

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Kernix 64 × 64 · f32	184	0.184
Create zeros / faer 64 × 64 · f32	328	0.328
Create zeros / ndarray 64 × 64 · f32	136	0.136
Create zeros / NumPy 64 × 64 · f32	512	0.512
Create zeros / Candle 64 × 64 · f32	256	0.256
Create ones / faer 64 × 64 · f32	239.99999999999997	0.23999999999999997
Create ones / Kernix 64 × 64 · f32	248	0.248
Create ones / Candle 64 × 64 · f32	344	0.344
Create ones / NumPy 64 × 64 · f32	1520	1.52
Create ones / ndarray 64 × 64 · f32	208	0.208